Muhammad Muizz Zafar

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Software Engineer | ETL Pipelines | Data Platforms | Cloud Infrastructure

Skills

Programming Languages:	Java, Python, TypeScript, GoLang, SQL, JavaScript, Kotlin, C#, C++, HTML/CSS, PHP
Tools/Frameworks:	AWS, AWS CDK, CloudFormation, Docker, Splunk, CloudWatch, Athena, Apache Spark, Amazon

Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - Dec 2024

Work experience

Software Developer — May 2024 - PRESENT
Elections Canada — Ottawa, ON

- Built automated **Python ETL validation pipelines** and **Azure DevOps CI/CD workflows** to verify production data movement across **20M + records**, across multiple environments reducing manual validation effort.
- Used **Apache Spark** to profile, transform, and validate datasets of up to **5+ GB** during pipeline verification, accelerating bulk data checks and anomaly detection across staging and production environments.
- Built internal **data analytics dashboards** using **Python, Streamlit, and Plotly Dash** to surface pipeline health, operational metrics, validation trends, and data quality issues for directorate and senior management.
- Coordinated version-controlled database deployments and release processes, improving traceability and reliability of **production data infrastructure** across shared environments.
- Collaborated in an Agile cross-functional team with cloud engineers and DevOps teams to improve **distributed data workflows**, release quality, and production reliability.

Full Stack Developer Intern — September 2023 - April 2024
University of Ottawa — Ottawa, ON

- Delivered production updates to the University of Ottawa emergency alert platform using **C#** and **Angular**, supporting a high-traffic service used by over **20,000** students daily.
- Designed and executed **unit and end-to-end tests** using **Jasmine**, **Cypress**, and **JUnit**, validating new frontend and backend features.
- Collaborated with engineering teams in an **Agile sprint cycle** to review test results, ensure feature reliability, and fix UI/API integration bugs.

Cloud Developer Intern — May 2023 - August 2023
Trend Micro Inc. — Ottawa, ON

- Developed serverless monitoring workflows using **AWS Lambda, S3, and CloudWatch** to analyze **50,000+ artifacts per day** and support automated vulnerability scanning pipelines for high-volume cloud artifacts.
- Defined and maintained cloud infrastructure using **Golang**, **AWS CDK**, through **CloudFormation** to provision **50+ AWS resources** and improve repeatability of service deployments.
- Implemented concurrent processing logic in **GoLang** using goroutines to improve throughput and strengthen fault handling for **distributed cloud data processing** workloads.
- Integrated vulnerability scanning tools (**Anchore, Grype**) with AWS services to automate security checks and operational monitoring across **50,000+ cloud artifacts per day**.

Cloud Developer Intern — September 2022 - December 2022
Trend Micro Inc. — Ottawa, ON

- Built a serverless automation pipeline using **AWS Lambda, S3, and Athena** to process incoming cloud data events, query metadata, and trigger downstream updates, reducing manual operational effort.
- Configured **Splunk** and **AWS CloudWatch** monitoring, alarms, dashboards, and alerts to track **15+ services** and improve observability of production cloud infrastructure.
- Developed **Docker-based integration tests** and unit tests to improve reliability of distributed serverless data pipelines.

Full Stack Developer Intern — January 2022 - April 2022
Applied Mind Inc. — Ottawa, ON

- Built **Docker** containers and automated event ingestion into **PostgreSQL** using **Python** and **GraphQl**, reducing testing time and improving deployment readiness.

Full Stack Developer — June 2019 - May 2021
Tech Under Twenty — Oakville, ON

- Developed and enhanced the club management web pages for admins by using bootstrap and CSS media queries in **Angular** which helped gain **30+** new users
- Developed RESTful APIs in **.Net Core (C#)** for updating users information that helped improve user satisfaction. Implemented CRUD APIs for user post system that increased userbase by **10%**